

New Media for a New Generation of Explorers

Realtime Underwater Modeling and Immersion

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1 INTRODUCTION

The Ocean Exploration Trust (OET) is a 501(c)(3) not-for-profit organization dedicated to exploring the ocean and seeking new discoveries in geology, biology, maritime history, archaeology, and chemistry while pushing the boundaries of education, outreach, and technological innovation.

OET's expeditions are launched aboard the Exploration Vessel (E/V) Nautilus—a 68-meter research vessel equipped with remotely operated vehicles (ROVs) reaching depths of up to 6000 meters. We collaborate with the broader research community to identify priority regions and phenomena and share our expeditions with explorers worldwide via live telepresence.

We are using 3D imagery to reach new audiences in new ways, allowing them to explore our work, our ship, and our educational resources as nonlinear 'explore-at-your-own-pace' experiences. We have previously worked with platforms like Matterport, Sketchfab, Nira.app, and Spatial.io, working with partners to bring these experiences to various audiences, whether focused on cultural storytelling, edutainment, classrooms, or at-home 3D printing.

The Realtime Underwater Modeling and Immersion Project (RUMI) is an Unreal Engine-based ROV/data simulator that enables dynamic and immersive exploration of our deep-sea exploration data. The simulation integrates 3D georeferenced models generated onboard in near real-time via photogrammetry with deep-sea bathymetry and various engineering/scientific metadata. This platform additionally incorporates our observations (video, photos, and scientific data logs), engineering, and environmental data, allowing us to visualize each dive's progression in 3D space, plot observations, display video used to generate 3D models, and graph sensor data across time.

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2 TECHNOLOGICAL INNOVATIONS / TESTING

RUMI required the development of new technologies for the ROVs, new data processes on the ship, and shoreside development of the visualization environment. These elements were integrated during a November 2023 cruise off the coast of Hawaii, NA156 "Exploration through Advanced Imaging".

2.1 Technological Innovations:

- **Widefield Camera Array (WCA) [1]:** The Widefield Camera Array was developed by the Ocean Exploration Trust and the Sexton Corporation with funding from the Office of Naval Research. It collects high-resolution imagery to produce rapid 3-dimensional photogrammetric models, hemispherical video for immersive imagery and virtual reality, and IMAX-quality cinematographic productions. The system consists of three genlocked E2-F6 cinema cameras that record synchronous images at extremely wide fields of view. Two cameras in the array operate in parallel to record stereoscopic images at a 180-degree angle of view, while the third camera captures a 60-107 degree image optimized for the light level, terrain, and altitude of a given survey. Each camera features a 24-megapixel full-frame sensor capable of imaging at 60 frames per second at 6064×2560 pixels per frame.
- **Integration of high-resolution multibeam echosounder bathymetry data:** We collected high-resolution bathymetric data with a NORBIT multibeam echosounder (MBES) and navigation solution fusing data from acoustic, and inertial navigation systems. The MBES emits acoustic pulses, generating soundings that, when combined with navigation data from the ROVs, produce a high-resolution digital model of the seafloor. The acoustic models are then fused with the photogrammetric 3D models to validate scale, geographic orientation and add additional context in areas not easily observed through optical means.

2.2 Data Processing Techniques:

- **Automated Model Generation:** Custom scripting was developed to process data from the Widefield Camera Array and ROV navigational data during deep-sea exploration dives. RealityCapture was used to process alignments as images were uploaded from the Widefield Camera Array from the ROV. Models were then generated on-ship, georeferenced with the ROV navigation data, and uploaded to Cesium Ion via an automated process for further processing ashore.



Figure 1: The Widefield Camera Array integrated on ROV Hercules. Three cameras make up the array, surveyed in relative to the NORBIT multibeam echosounder (top-right).

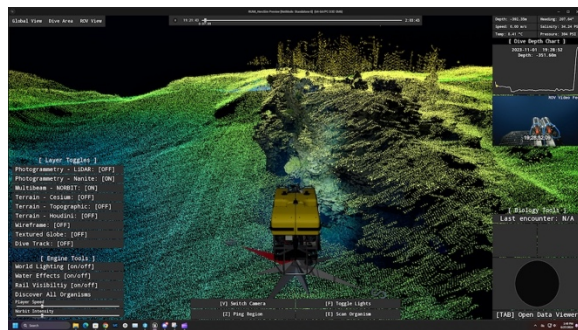


Figure 2: The Realtime Underwater Modeling and Immersion data simulation environment, integrating data sources from the ROV navigation, multibeam, and Widefield Camera Array 3D models.

- **Unified 3D Exploration Platform [2]:** An Unreal Engine-based platform was developed to integrate these data in 3D space and time, incorporating the georeferenced models, high-resolution multibeam bathymetry, scientific observations, and video streams into a single unified spatial-temporal snapshot for each dive. The simulator was built with future expansion in mind for both scientific and educational purposes, focusing on developing sustainable pipelines for the rapid integration of new data generated during future dives. This platform transforms disparate data streams into cohesive 3D space, contextualizing deep ocean observations, providing new ways to visualize data over space and how they change over time. In this ways, the platform creates an immersive experience that could be further developed as a stand-alone ‘game’ to teach about deep ocean exploration, discoveries, and technologies.

3 CONCLUSION

The Realtime Underwater Modeling and Immersion Project (RUMI) demonstrates a new paradigm of how deep-sea exploration data can be integrated, visualized, and shared. By integrating advanced imaging technologies such as the Widefield Camera Array and the NORBIT multibeam echosounder with a sophisticated Unreal Engine-based simulation platform, RUMI transforms traditional linear data streams into dynamic, interactive 3D experiences. Looking ahead, the RUMI project sets a robust foundation for future advancements in oceanographic data visualization.

The platform’s design allows for continuous expansion and adaptation, ensuring it can accommodate new data and technological developments. By automating the data processing pipeline and leveraging widely-supported tools such as RealityCapture and Cesium Ion, RUMI ensures the sustainability and scalability of its methodologies. We demonstrated the capacity of this process to visualize data in 3D in near-real-time during active exploration activities. The successful implementation of RUMI during these activities highlights its potential to revolutionize deep-sea data visualization and exploration, paving the way for future innovations and discoveries in marine science.